

EXPERIMENT – 08

Experiment No.	08
Date	_____
Title	Interfaces – Employee Monthly Salary Details

1. Objective / Aim

To calculate and display the monthly salary details of regular and contract employees using a Java interface.

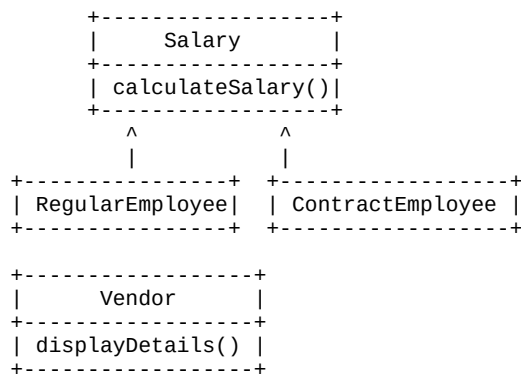
2. System Requirements

Hardware	Computer/Laptop with keyboard and monitor
Software	JDK (Java Development Kit), Command Prompt or any Java IDE

3. Algorithm / Logic Description

1. Define a Salary interface with calculateSalary().
2. Implement the interface in RegularEmployee and ContractEmployee classes.
3. For regular employees, calculate total salary as Basic Pay + HRA + TA = 25000 + 15000 + 5000 = 45000.
4. For contract employees, calculate total salary as Basic Pay + HRA + TA = 12000 + 0 + 3000 = 15000.
5. Read the employee ID from the user. IDs starting with R indicate regular employees, C indicate contract employees, and V indicate vendors.
6. Display the matching salary details. Vendor salary is not provided in the question, so display a message for vendors.
7. Stop the program.

4. UML Class Diagram



5. Source Code

```
import java.util.Scanner;

interface Salary {
    void calculateSalary();
}

class RegularEmployee implements Salary {
    String employeeId;
    int basicPay = 25000;
    int hra = 15000;
    int ta = 5000;

    RegularEmployee(String employeeId) {
        this.employeeId = employeeId;
    }

    public void calculateSalary() {
        int total = basicPay + hra + ta;
    }
}
```

```

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("HRA: " + hra);
        System.out.println("T.A: " + ta);
        System.out.println("Total Amount: " + total);
    }
}

class ContractEmployee implements Salary {
    String employeeId;
    int basicPay = 12000;
    int hra = 0;
    int ta = 3000;

    ContractEmployee(String employeeId) {
        this.employeeId = employeeId;
    }

    public void calculateSalary() {
        int total = basicPay + hra + ta;

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("HRA: " + hra);
        System.out.println("T.A: " + ta);
        System.out.println("Total Amount: " + total);
    }
}

class Vendor {
    String vendorId;

    Vendor(String vendorId) {
        this.vendorId = vendorId;
    }

    void displayDetails() {
        System.out.println("Vendor ID: " + vendorId);
        System.out.println("Vendor salary details are not specified in the question.");
    }
}

public class EmployeeSalary {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee Id: ");
        String id = sc.nextLine().trim().toUpperCase();

        if (id.startsWith("R")) {
            Salary employee = new RegularEmployee(id);
            employee.calculateSalary();
        } else if (id.startsWith("C")) {
            Salary employee = new ContractEmployee(id);
            employee.calculateSalary();
        } else if (id.startsWith("V")) {
            Vendor vendor = new Vendor(id);
            vendor.displayDetails();
        } else {
            System.out.println("Invalid Employee ID.");
            System.out.println("Use R for Regular, C for Contract, or V for Vendor.");
        }

        sc.close();
    }
}

```

6. Compilation & Execution Commands

```

javac EmployeeSalary.java
java EmployeeSalary

```

7. Output Screen

Test Case 1 – Regular Employee

```

Enter Employee Id: R101
Salary Details:
Basic Pay: 25000
HRA: 15000
T.A: 5000
Total Amount: 45000

```

Test Case 2 – Contract Employee

```

Enter Employee Id: C101
Salary Details:

```

Basic Pay: 12000
HRA: 0
T.A: 3000
Total Amount: 15000

Test Case 3 – Vendor

Enter Employee Id: V101
Vendor ID: V101
Vendor salary details are not specified in the question.

Test Case 4 – Invalid ID

Enter Employee Id: X101
Invalid Employee ID.
Use R for Regular, C for Contract, or V for Vendor.

8. Result

Thus, the Java program using an interface to calculate and display monthly salary details for regular and contract employees was implemented and executed successfully. Vendor details are handled separately because vendor salary values were not specified in the given problem.